

The r -wreath extension of Theorem D: S_r -isotypic decomposition of $R_{(k^r)}$ and a cyclotomic Hilbert-series identity

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Abstract

Fix integers $r, k \geq 1$, let $n = rk$, and let $\alpha = (k^r)$ be the composition with r parts each equal to k . Write $S_\alpha = S_k^r \subseteq S_n$ for the Young subgroup and $H = S_k \wr S_r$ for its wreath normaliser. The residual quotient $H/S_\alpha \cong S_r$ acts on the parabolic coinvariant algebra $R_\alpha = R_n^{S_\alpha}$ by graded ring automorphisms. We prove:

- (A) the graded S_r -isotypic multiplicities are $m_\pi(t) = \sum_\lambda \tilde{f}_\lambda(t) \cdot \langle s_\pi[h_k], s_\lambda \rangle$;
- (B) the graded Frobenius character is $\text{Frob}_t(R_\alpha) = \sum_\pi m_\pi(t) s_\pi$;
- (C) at $k = 2$, the total Hilbert series admits the cyclotomic product factorisation $\binom{2r}{2r}_t = [r]_{t^2}! \cdot \prod_{i=1}^r [2i-1]_t$.

Result (A)+(B) extends the $r = 2$ Theorem D of *Wreath decomposition of the collision-regime coset Poincaré identity* (2026-07-30) to arbitrary r ; result (C) is a clean closed form for the “collision regime” Hilbert series at $k = 2$. We furthermore verify computationally that at $r = 3, k = 2$ the cyclotomic factorisation of (C) is *not* aligned with the isotypic decomposition of (A): the factor $\Phi_3 = [3]_t$ divides $\binom{6}{2,2,2}_t$ but no individual $m_\pi(t)$.

1 Setup

Let S_n act on the polynomial ring $\mathbb{C}[x_1, \dots, x_n]$ by permutation. Let $I_n = (\mathbb{C}[x_1, \dots, x_n]_+^{S_n})$ be the ideal generated by the positive-degree symmetric polynomials, and set

$$R_n = \mathbb{C}[x_1, \dots, x_n]/I_n,$$

the *coinvariant algebra*, a graded S_n -module with total Hilbert series $[n]_t!$. By Chevalley–Shephard–Todd (and Steinberg for the grading),

$$R_n \cong \bigoplus_{\lambda \vdash n} \tilde{f}_\lambda(t) \cdot V_\lambda \quad \text{as graded } S_n\text{-modules,}$$

where V_λ is the irreducible S_n -module of shape λ and $\tilde{f}_\lambda(t)$ is its *fake degree*. In terms of hook lengths, $\tilde{f}_\lambda(t) = t^{n(\lambda)} [n]_t! / \prod_{c \in \lambda} [h(c)]_t$ where $n(\lambda) = \sum_i (i-1)\lambda_i$.

Fix a composition $\alpha = (\alpha_1, \dots, \alpha_r)$ of n , and let $S_\alpha = S_{\alpha_1} \times \dots \times S_{\alpha_r} \subseteq S_n$ be the corresponding Young subgroup. The *parabolic coinvariant algebra* is

$$R_\alpha := R_n^{S_\alpha}.$$

Its Hilbert series is the q -multinomial

$$\dim_t R_\alpha = \binom{n}{\alpha_1, \dots, \alpha_r}_t = \frac{[n]_t!}{[\alpha_1]_t! \cdots [\alpha_r]_t!}.$$

(Reference: Garsia–Stanton, JLS 1984; also follows from $\dim_t R_\alpha = \sum_\lambda \tilde{f}_\lambda(t) \langle s_\lambda, h_\alpha \rangle$ and the Cauchy identity $\sum_\lambda s_\lambda h_\alpha = h_\alpha$.)

Specialise now to $\alpha = (k^r)$, i.e., all parts equal to k . The wreath product

$$H := S_k \wr S_r = S_\alpha \rtimes S_r$$

is the normaliser of S_α in S_{rk} ; the projection $H \twoheadrightarrow S_r$ has kernel S_α .

2 Main theorem

Theorem 1. *Let $r, k \geq 1$, $n = rk$, $\alpha = (k^r)$, and $H = S_k \wr S_r$.*

- (i) *H preserves $R_\alpha = R_n^{S_\alpha}$ and S_α acts trivially, so $H/S_\alpha \cong S_r$ acts on R_α by graded ring automorphisms.*
- (ii) *The graded Frobenius character of R_α as an S_r -module is*

$$\text{Frob}_t(R_\alpha) = \sum_{\pi \vdash r} m_\pi(t) s_\pi,$$

with graded isotypic multiplicities

$$m_\pi(t) = \sum_{\lambda \vdash n} \tilde{f}_\lambda(t) \cdot \langle s_\pi[h_k], s_\lambda \rangle.$$

- (iii) *The total Hilbert series is*

$$\dim_t R_\alpha = \sum_{\pi \vdash r} f^\pi m_\pi(t) = \binom{n}{k, k, \dots, k}_t,$$

where $f^\pi = \dim V_\pi$.

Proof. (i) The subgroup $H \subseteq S_n$ normalises S_α , so H acts on the S_α -invariants $R_n^{S_\alpha} = R_\alpha$, and S_α acts trivially on its own invariants. Consequently the action factors through $H/S_\alpha \cong S_r$.

(ii) Applying the exact functor $(-)^{S_\alpha}$ (since we are in characteristic zero) to the CST decomposition,

$$R_\alpha = \bigoplus_{\lambda \vdash n} \tilde{f}_\lambda(t) \cdot V_\lambda^{S_\alpha} \quad \text{as graded } S_r\text{-modules,}$$

where the S_r -action on each $V_\lambda^{S_\alpha}$ comes from the H -action on $V_\lambda|_H$ (well-defined by the same normalisation argument).

Fix $\pi \vdash r$. Let $\tilde{V}_\pi := V_\pi \boxtimes \text{triv}^{\otimes r}$ denote the H -module obtained by pulling back V_π along $H \twoheadrightarrow S_r$; equivalently, S_α acts trivially and S_r acts by V_π . Then:

$$\begin{aligned} \text{Hom}_{S_r}(V_\pi, V_\lambda^{S_\alpha}) &= \text{Hom}_H(\tilde{V}_\pi, V_\lambda^{S_\alpha}) && (S_\alpha \text{ trivial on both}) \\ &= \text{Hom}_H(\tilde{V}_\pi, V_\lambda|_H) && (\text{image is } S_\alpha\text{-fixed}) \\ &= \text{Hom}_{S_n}(\text{Ind}_H^{S_n}(\tilde{V}_\pi), V_\lambda). && (\text{Frobenius reciprocity}) \end{aligned}$$

The Frobenius character of the induced S_n -module $\text{Ind}_H^{S_n}(\tilde{V}_\pi)$ is the plethysm $s_\pi[h_k]$: this is the classical formula for wreath induction of a representation trivial on the base subgroup (Macdonald, *Symmetric Functions and Hall Polynomials*, Ch. I, App. A, eq. (A.11); Stanley, *EC2*, ex. 7.75). Hence

$$\dim \text{Hom}_{S_r}(V_\pi, V_\lambda^{S_\alpha}) = \langle s_\pi[h_k], s_\lambda \rangle.$$

Summing λ with graded weight $\tilde{f}_\lambda(t)$:

$$m_\pi(t) := \dim_t \text{Hom}_{S_r}(V_\pi, R_\alpha) = \sum_\lambda \tilde{f}_\lambda(t) \langle s_\pi[h_k], s_\lambda \rangle.$$

(iii) Direct consequence of $R_\alpha \cong \bigoplus_\pi m_\pi(t) \cdot V_\pi$ and $\dim V_\pi = f^\pi$; equality with the q -multinomial is the classical Hilbert-series formula for parabolic coinvariants recalled in §1. $\square$

3 Cyclotomic factorisation at $k = 2$

Theorem 2. *For every $r \geq 1$,*

$$\binom{2r}{2, 2, \dots, 2}_t = [r]_{t^2}! \cdot \prod_{i=1}^r [2i-1]_t.$$

Proof. Since $[1]_t! = [2]_t$ (with $[2]_t = 1+t$), we have $[2]_t! = [2]_t$ and $([2]_t!)^r = [2]_t^r$. Thus

$$\binom{2r}{2^r}_t = \frac{[2r]_t!}{[2]_t^r}.$$

Split $[2r]_t! = \prod_{m=1}^{2r} [m]_t$ into even and odd factors:

$$[2r]_t! = \left(\prod_{i=1}^r [2i]_t \right) \cdot \left(\prod_{i=1}^r [2i-1]_t \right).$$

The elementary identity

$$[2i]_t = [2]_t \cdot [i]_{t^2} \quad \left(\text{since } \frac{1-t^{2i}}{1-t} = \frac{1-t^2}{1-t} \cdot \frac{1-t^{2i}}{1-t^2} \right)$$

gives $\prod_{i=1}^r [2i]_t = [2]_t^r \cdot [r]_{t^2}!$. Substituting and cancelling $[2]_t^r$:

$$\binom{2r}{2^r}_t = [r]_{t^2}! \cdot \prod_{i=1}^r [2i-1]_t. \quad \square$$

Remark (Combinatorial interpretation). The specialisation $t = 1$ recovers $\binom{2r}{2r} = r! \cdot (2r - 1)!!$, the standard split of ordered pair-decompositions of $\{1, \dots, 2r\}$ into (matching selection) $\times$ (pair ordering). The factor $[r]_{t^2}!$ is the pair-ordering contribution in t^2 : swapping two adjacent pairs contributes 2 to the major index (each pair is stored increasingly, so a swap crosses each side of the other pair once).

Remark (Cyclotomic vs. isotypic decomposition). The factorisation of Theorem ?? is a *Hilbert-series* identity and does not, in general, align with the S_r -isotypic decomposition of Theorem ??. For $(r, k) = (3, 2)$ we verify computationally (§4) that $[3]_t = \Phi_3$ divides $\binom{6}{2,2,2}_t = [5]_t [3]_t [3]_{t^2} [2]_{t^2}$ but does *not* divide any individual $m_\pi(t)$. In contrast, the factor $[5]_t = [2r - 1]_t$ turns out to divide *every* $m_\pi(t)$; this divisibility appears empirically to hold whenever $k = 2$ (verified $r = 2, 3, 4$) and additionally at some higher (r, k) (e.g. $(3, 3)$ but not $(2, 3)$), but does not follow from the framework of this note.

4 Verification at $(r, k) = (3, 2)$

The plethysms $s_\pi[h_2]$ for $\pi \vdash 3$ are (computed in `probes/2026-07-30-r3-wreath-extension/plethysm.py`):

$$\begin{aligned} s_{(3)}[h_2] &= s_{(6)} + s_{(4,2)} + s_{(2,2,2)}, \\ s_{(2,1)}[h_2] &= s_{(5,1)} + s_{(4,2)} + s_{(3,2,1)}, \\ s_{(1,1,1)}[h_2] &= s_{(4,1,1)} + s_{(3,3)}. \end{aligned}$$

The first is the classical Foulkes formula for $h_3[h_2]$. Summed with S_3 -dimension weights ($f^{(3)} = 1$, $f^{(2,1)} = 2$, $f^{(1^3)} = 1$):

$$s_{(3)}[h_2] + 2s_{(2,1)}[h_2] + s_{(1,1,1)}[h_2] = s_{(6)} + 2s_{(5,1)} + 3s_{(4,2)} + s_{(4,1,1)} + s_{(3,3)} + 2s_{(3,2,1)} + s_{(2,2,2)} = h_2^3,$$

consistent with $\text{Frob}(\text{Ind}_{S_\alpha}^{S_6}(\text{triv})) = h_2^3$.

The fake degrees of the six relevant shapes (via $\tilde{f}_\lambda(t) = t^{n(\lambda)}[6]_t! / \prod [h(c)]_t$):

λ	$\tilde{f}_\lambda(t)$
(6)	1
(5, 1)	$t + t^2 + t^3 + t^4 + t^5$
(4, 2)	$t^2 + t^3 + 2t^4 + t^5 + 2t^6 + t^7 + t^8$
(4, 1, 1)	$t^3 + t^4 + 2t^5 + 2t^6 + 2t^7 + t^8 + t^9$
(3, 3)	$t^3 + t^5 + t^6 + t^7 + t^9$
(3, 2, 1)	$t^4 + 2t^5 + 2t^6 + 3t^7 + 3t^8 + 2t^9 + 2t^{10} + t^{11}$
(2, 2, 2)	$t^6 + t^8 + t^9 + t^{10} + t^{12}$

Applying Theorem ??(ii):

$$\begin{aligned} m_{(3)}(t) &= \tilde{f}_{(6)} + \tilde{f}_{(4,2)} + \tilde{f}_{(2,2,2)} \\ &= 1 + t^2 + t^3 + 2t^4 + t^5 + 3t^6 + t^7 + 2t^8 + t^9 + t^{10} + t^{12}, \\ m_{(2,1)}(t) &= \tilde{f}_{(5,1)} + \tilde{f}_{(4,2)} + \tilde{f}_{(3,2,1)} \\ &= t + 2t^2 + 2t^3 + 4t^4 + 4t^5 + 4t^6 + 4t^7 + 4t^8 + 2t^9 + 2t^{10} + t^{11}, \\ m_{(1,1,1)}(t) &= \tilde{f}_{(4,1,1)} + \tilde{f}_{(3,3)} \\ &= 2t^3 + t^4 + 3t^5 + 3t^6 + 3t^7 + t^8 + 2t^9. \end{aligned}$$

Weighted sum:

$$\begin{aligned}
m_{(3)}(t) + 2m_{(2,1)}(t) + m_{(1,1,1)}(t) &= 1 + 2t + 5t^2 + 7t^3 + 11t^4 + 12t^5 + 14t^6 \\
&\quad + 12t^7 + 11t^8 + 7t^9 + 5t^{10} + 2t^{11} + t^{12} \\
&= \binom{6}{2,2,2}_t \\
&= [5]_t \cdot [3]_t \cdot [3]_{t^2} \cdot [2]_{t^2}. \quad \checkmark
\end{aligned}$$

Palindromicity of each $m_\pi(t)$ is inherited from the palindromicity of every $\tilde{f}_\lambda(t)$ (the coinvariant algebra is a Frobenius algebra with top degree $\binom{n}{2}$; for $n = 6$, top degree 15). The individual multiplicities are centred at degrees 6, 6, 6 respectively (their supports are $[0, 12]$, $[1, 11]$, $[3, 9]$).

5 Computational verification

The identity of Theorem ??(iii),

$$\sum_{\pi \vdash r} f^\pi m_\pi(t) = \binom{rk}{kr}_t,$$

has been verified in `probes/2026-07-30-r3-wreath-extension/verify_full.py` for

$$(r, k) \in \{(2, 2), (2, 3), (3, 2), (2, 4), (3, 3), (4, 2)\}.$$

The cyclotomic identity of Theorem ?? was verified symbolically for $r = 1, 2, \dots, 6$ in the same script.

6 Remarks and follow-up questions

Remark (Relation to Levicán–Romero). The framework of Levicán–Romero (arXiv:2504.19008) develops $\mathbb{Z}_k \wr S_n$ -class functions for wreath Macdonald theory. The present setup ($S_k \wr S_r$ acting on $R_{rk}^{S_r}$) is classical rather than wreath-Macdonald, but the underlying combinatorics (plethysm $s_\pi[h_k]$, Frobenius reciprocity for H) is a degenerate specialisation of their machinery. A bigraded (q, t) -lift following Szendrői arXiv:2602.15017 or Romero–Wen arXiv:2505.01732 is the natural next question.

Remark ($[2r - 1]_t$ -divisibility of $m_\pi(t)$). Empirically, $[2r - 1]_t$ divides every $m_\pi(t)$ when $k = 2$ (verified $r = 2, 3, 4$), and also at $(r, k) = (3, 3)$, but does not at $(2, 3)$ or $(2, 4)$. This suggests a rank-two phenomenon governed by the divisibility of the fake degrees $\tilde{f}_\lambda(t)$ by cyclotomic factors indexed by hook data. A conceptual explanation (via, e.g., Springer-fibre cohomology or Solomon’s descent-algebra picture) is not attempted here.

Remark (Sprint context). This note completes Theorem D of the byproduct paper *Modified Hall–Littlewood boundary* (§5.2) at general r . The $r = 2$ case was proved in the companion note *Wreath decomposition of the collision-regime coset Poincaré identity* (2026-07-30, morning); this extension supplies the general- r version of Conjectures A, B, and (partially) C from `PROVE.md` of 2026-07-30 late-wake. Conjecture C is proved as a Hilbert-series identity but shown to be *not* the S_r -isotypic decomposition.